

NATHANAEL JO

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Research Interests	AI Alignment; Machine Learning; Economics and Computation; Human-AI Interaction	
Academics	Massachusetts Institute of Technology , Cambridge, MA	2023 – Present
	PhD Computer Science	GPA: 4.80/5.00
	<i>Research Advisors:</i> Manish Raghavan and Ashia Wilson <i>Relevant Coursework:</i> Optimization Methods, Probability, Deep Learning, Market Design	
	University of Southern California , Los Angeles, CA	2017 – 2021
	MS Data Science	GPA: 3.90/4.00
	BA Applied Mathematics	GPA: 3.98/4.00
	<i>Relevant Coursework:</i> Machine Learning, Algorithms, Data Mining, Knowledge Graphs, Numerical Methods, Probability & Statistics, Real Analysis, Data Visualization	
Grants, Awards, and Honors	Herbert E. Grier (1933) and Dorothy J. Grier Fellowship	2023 – 2024
	USC Discovery Scholar Prize	2021
	Epstein Industrial and Systems Engineering Fellowship	2021
	USC Viterbi School of Engineering Anonymous Endowment	2021
	Presidential Scholarship	2017 – 2021
	Harry A. Miller Endowed Scholarship	2019 – 2020
	Academic Achievement Award	2018 – 2020
	USC Undergraduate Research Associate Grant	2018
	Phi Beta Kappa Honor Society	
Journal Papers	Published Papers	
	[1] “Learning Optimal Prescriptive Trees from Observational Data,” N. Jo , S. Aghaei, A. Gómez, P. Vayanos. <i>Management Science</i> , 2026.	
	• <u>INFORMS Undergraduate Operations Research Prize Award 2021, Finalist</u>	
	[2] “Drop a Line, Submit on Time? Experimental Evidence on the Effect of Tailored Predeadline Reminders on Pollution Reporting,” E. Benami, N. Jo , D. Ho. <i>Journal of the Association of Environmental and Resource Economists</i> , 2026.	
	[3] “Not (Officially) in My Backyard: Characterizing Informal Accessory Dwelling Units and Informing Housing Policy with Remote Sensing,” N. Jo* , A. Vallebuena*, D. Ouyang, D. Ho. <i>Journal of the American Planning Association</i> , 2024.	
	Papers Under Review	
	[4] “ODTLearn: A Python Package for Learning Optimal Decision Trees for Prediction and Prescription,” P. Vossler, S. Aghaei, N. Justin, N. Jo , P. Vayanos, A. Gómez. Submitted, 2025.	

Conference Papers

Published Papers

- [5] “Position: AI Evaluations Should be Grounded on a Theory of Capability,” **N. Jo**, A. Wilson. *International Conference for Machine Learning (ICML)*, 2026.
- [6] “Homogeneous Algorithms Can Reduce Competition in Personalized Pricing,” **N. Jo** K. Creel, A. Wilson, M. Raghavan. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- Also in *International Conference on Machine Learning (ICML)*, Workshop on Humans, Algorithmic Decision-Making, and Society, 2024.
 - Also in *Conference on Neural Information Processing Systems (NeurIPS)*, Workshop on Regulatable ML, 2024.
- [7] “Fairness in contextual resource allocation systems: metrics and incompatibility results,” **N. Jo***, B. Tang*, K. Dullerud, S. Aghaei, P. Vayanos. *AAAI Conference on Artificial Intelligence* 2023.
- [8] “Learning Fair Optimal Classification Trees: Trade-offs Between Interpretability, Fairness, and Accuracy,” **N. Jo**, S. Aghaei, J. Benson, A. Gómez, P. Vayanos. *AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES)*. 2023.
- [9] “Estimating and Implementing Fairness Metrics with Probabilistic Protected Features,” H. Elzayn , E. Black, P. Vossler, **N. Jo**, J. Goldin, D. Ho. *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2024.

Papers Under Review

- [10] “The Subjectivity of Monoculture,” **N. Jo**, N. Garg, M. Raghavan. 2026.
- [11] “Position: Alignment has a Fantasia Problem,” **N. Jo***, Z. Simone, M. Gordon, A. Wilson. 2026.

Talks

Conference Presentations

- “Learning Optimal Prescriptive Trees from Observational Data”
- INFORMS Annual Conference 2022: Session on Discrete Optimization for Society and Technology
 - INFORMS Annual Conference 2021: Session on Interpretable Machine Learning – Exact and Approximation Algorithms
 - CORS Annual Conference 2021: Session on ML/OR for Social Good
- “Learning Optimal Fair Decision Trees”
- INFORMS Annual Conference 2022: Session on Interpretable Machine Learning for Social Good
- “Fairness in contextual resource allocation systems: metrics and incompatibility results”
- INFORMS Annual Conference 2022: Session on Fairness and Equity in Operations

Other Research Experience	Regulation, Evaluation, and Governance Lab , Stanford University <i>Computational Research Fellow</i> , Principal Investigator: Dan E. Ho	2021 – 2023
	USC Center for Artificial Intelligence in Society <i>Affiliate Researcher</i> , Principal Investigator: Phebe Vayanos	2020 – 2023
	Security and Political Economy Lab , USC <u>Principal Investigator</u> : Benjamin Graham	2018 – 2020
	Performed data cleanup, analysis, and visualization in R on a 40-variable timeseries dataset with 180 countries to evaluate trends in nations' laws and its correlation to how governments distribute power	
Professional Experience	Africa Regional Grant on HIV , UNDP & USC Program on Global Health <u>Principal Investigator</u> : Laura Ferguson	2018
	Analyzed the progress of sub-Saharan African countries in removing the legal barriers for vulnerable populations living with HIV: an endline evaluation of a United Nations grant	
	Goldman Sachs , New York City <i>Quantitative Strats Intern</i>	2026
	- Built an evaluation and reporting framework for 8 internal credit risk ML models, integrating data engineering, model validation, and rigorous statistical analyses into a fully automated quarterly workflow - Developed an LLM-powered interpretation layer that identifies concerning test results and generates actionable diagnostics, improving the transparency and efficiency of model review	
	OutRight Action International , New York City <i>UN Program Intern</i>	2021
	- Monitoring developments at the UN regarding LGBTQI+ issues and drafting communications to governments and UN bodies - Conducting legal research to support civil society advocacy efforts, particularly to advance LGBTQI+ interests	
	Sony Pictures Entertainment , Los Angeles <i>Data Science Intern</i>	2019
Teaching Experience	- Revamped various Agile metrics for 80 teams in a pilot project to drive increased operational efficiency - Evaluated competing models that predict upcoming movie earnings to optimize resource allocation	
	University of Southern California <i>Teaching Assistant</i>	2018 – 2020
	MATH226 Multivariable Calculus (undergraduate level, ~200 students)	

Activities and Involvements	MIT EECS Graduate Application Assistance Program (GAAP)	2023
	USC Kappa Sigma Fraternity <i>VP of Philanthropy, VP of External Relations</i>	2018 – 2021
	USC Queer & Ally Student Assembly <i>Greek Life Student Liaison</i>	2018 – 2021
	Joint Educational Program (USC) <i>Volunteer Teacher</i> • Taught French to 30 LAUSD students twice a week as part of a volunteer program	2019
Languages	English and Indonesian (native), French (intermediate), Mandarin (basic)	
Technical Skills	• Building, finetuning, and auditing machine learning/AI models in Python using Hugging Face, PyTorch, Tensorflow, scikit, matplotlib, etc. • Computer vision tools and frameworks (PIL, Rasterio, OpenCV, etc.) • NLP tools and frameworks (Transformers, NLTK, spaCy, etc.) • Optimization and quantitative modelling • Large-scale parallel computing using Apache Hadoop, Spark, and Dask • Data management (MongoDB, Apache Suite, etc.) and database modelling • Other languages: R, SQL, C/C++, Java, Swift	